

## Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 6: Solving Right-Angled Triangles Using Trigonometric Ratios

Teacher reference only. Do not project or hand out.

### Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 |
|----|----|----|----|----|----|----|----|
| A  | A  | A  | B  | A  | D  | A  | A  |

USE AFTER: PREDICT PHASE — Predict: Diagram A, the Kisumu County Referral Hospital ramp (choosing the ratio)

**Q1. A ramp at Kisumu County Referral Hospital makes a  $12^\circ$  angle with the ground. The horizontal distance along the ground is 6 m. You want the length of the ramp surface. Which ratio should you use?**

- A. Cosine, because the known side is adjacent to  $12^\circ$  and the ramp is the hypotenuse**
- B. Sine, because the ramp is the side opposite the  $12^\circ$  angle
- C. Tangent, because the ground distance is the side opposite the  $12^\circ$  angle
- D. Tangent, because a ramp is a slope

**Correct answer: A.** The 6 m ground distance is adjacent to the  $12^\circ$  angle and the ramp surface is the hypotenuse. Adjacent and hypotenuse means CAH, so cosine is used.

USE AFTER: PREDICT PHASE — Predict: Diagram C, the Garissa telecom mast support wire

**Q2. A telecom mast in Garissa is 45 m tall and stands at a right angle to the ground. A support wire is anchored on the ground 45 m from the base of the mast. What angle does the wire make with the ground?**

- A.  $45^\circ$**
- B.  $30^\circ$
- C.  $60^\circ$
- D.  $90^\circ$

**Correct answer: A.** The known sides are opposite (45 m) and adjacent (45 m), so  $\tan \theta = 45/45 = 1$ . The angle whose tangent is 1 is  $45^\circ$ , a special angle.

USE AFTER: OBSERVE PHASE — Observe: worked example 'WORKED EXAMPLE — Nakuru Mast'

**Q3. A student stands 40 m from the base of the Nakuru transmission mast. The angle of elevation to the top is  $50^\circ$ . Using  $\tan 50^\circ \approx 1.1918$ , what is the height of the mast?**

- A. 47.7 m**
- B. 33.6 m
- C. 30.6 m
- D. 25.7 m

**Correct answer: A.** The known side (40 m) is adjacent and the height is opposite, so  $\tan 50^\circ = \text{height}/40$ . Then height =  $40 \times 1.1918 \approx 47.7$  m.

USE AFTER: OBSERVE PHASE — Observe: second demonstration, the Kisumu ramp co-constructed with the teacher

**Q4. A ramp makes a  $12^\circ$  angle with the ground. The horizontal distance along the ground is 6 m. Using  $\cos 12^\circ \approx 0.9781$ , what is the length of the ramp surface, to 2 decimal places?**

- A. 1.28 m
- B. 6.13 m**
- C. 28.88 m
- D. 5.87 m

**Correct answer: B.**  $\cos 12^\circ = \text{adjacent}/\text{hypotenuse} = 6/\text{ramp}$ , so ramp =  $6 \div 0.9781 \approx 6.13$  m. Multiplying by  $\cos 12^\circ$  is a rearrangement error and gives a value shorter than 6 m, which is impossible for a hypotenuse.

USE AFTER: EXPLAIN PHASE — Explain: pair work, Problem 1 (Mombasa port ramp)

**Q5. A ramp at Mombasa port makes a  $15^\circ$  angle with the horizontal. The ramp is 8 m long. What is the vertical height the ramp rises, to 2 decimal places?**

- A. 2.07 m**
- B. 2.14 m
- C. 30.90 m
- D. 7.73 m

**Correct answer: A.** The height is opposite  $15^\circ$  and the ramp is the hypotenuse, so  $\sin 15^\circ = \text{height}/8$ . Height  $= 8 \times \sin 15^\circ \approx 8 \times 0.2588 \approx 2.07$  m.

USE AFTER: EXPLAIN PHASE — Explain: pair work, Problem 3 (Garissa tower shadow) and the link back to the Nakuru mast

**Q6. A surveyor in Garissa measures that a telecom tower casts a shadow 32 m long on flat ground when the sun's angle of elevation is  $62^\circ$ . What is the height of the tower, to 1 decimal place?**

- A. 17.0 m
- B. 28.3 m
- C. 15.0 m
- D. 60.2 m**

**Correct answer: D.** The shadow (32 m) is adjacent to  $62^\circ$  and the tower height is opposite, so  $\tan 62^\circ = \text{height}/32$ . Height  $= 32 \times 1.8807 \approx 60.2$  m.

USE AFTER: DQB PHASE — Driving Question Board: 'What does solving a right-angled triangle completely mean — and can we now do it?'

**Q7. After solving a right-angled triangle, a learner writes: 'I know one acute angle and one side. Now I can find the other two sides and the other angle.' What does 'solving a right-angled triangle completely' mean?**

- A. Finding every unknown side and angle from one known side and one acute angle**
- B. Finding the height of the triangle only
- C. Finding only the hypotenuse
- D. Finding the three ratios sin, cos and tan for the  $90^\circ$  angle

**Correct answer: A.** Solving completely means finding all unknown sides and angles. With one side and one acute angle, the ratios give the other sides, and the angle sum of  $180^\circ$  gives the remaining angle.

USE AFTER: MODEL BUILDING PHASE — Model: writing the decision rule box (SOH if O and H, CAH if A and H, TOA if O and A)

**Q8. For a named acute angle in a right-angled triangle, you know the length of the side opposite it and you want the length of the hypotenuse. Which rule from SOH-CAH-TOA should you use?**

- A. SOH, because it uses opposite and hypotenuse**
- B. CAH, because it uses the adjacent side
- C. TOA, because it uses the opposite side
- D. TOA, because tangent is the ratio to use whenever an angle is given

**Correct answer: A.** Opposite and hypotenuse are involved, so SOH applies:  $\sin \theta = \text{opposite}/\text{hypotenuse}$ . Tangent only links opposite and adjacent.